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Early-Stage Cardiovascular Disease Prediction Using a Late-Fusion Integration of Clinical and Lifestyle Data with Explainable Machine Learning

Machine Learning System

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Declaration

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Abstract

Non-communicable diseases now account for at least 43 million deaths each year, approximately 75% of all non-pandemic deaths worldwide, and cardiovascular disease alone is responsible for at least 19 million of them (World Health Organization, 2025). In the United Kingdom, cardiovascular disease imposed an estimated economic burden of £29.0 billion in 2021/22, of which £16.6 billion fell directly on health and care services (Shih *et al.*, 2025), while diabetes cost the UK almost £14 billion over the same period (Hex *et al.*, 2024). A substantial proportion of this burden is a consequence of late detection: 41.3% of common cancers in England were still diagnosed at stage 3 or 4 in the year to August 2024 (NHS England, 2025), and an estimated 1.3 million people in the UK are living with undiagnosed type 2 diabetes (Diabetes UK, 2025).

This study develops and evaluates a hybrid, explainable machine learning framework for the early-stage prediction of cardiovascular disease, with type 2 diabetes used as a secondary target to test whether the pipeline transfers across conditions. The contribution is deliberately narrower and more precise than "combining clinical and lifestyle data", a practice that is already well established in health informatics. Instead, the framework advances four specific and testable design decisions: (i) a dual-branch *late-fusion* architecture in which clinical and lifestyle modalities are modelled separately and combined through a calibrated meta-learner, rather than the near-universal early fusion by column concatenation; (ii) explicitly engineered clinical × lifestyle interaction features that encode known pathophysiological effect modification; (iii) missingness-aware treatment of self-reported behavioural data, in which missingness indicators are retained as informative features and imputation strategy is treated as an experimental variable rather than an assumption; and (iv) a strict ablation benchmark in which a clinical-only model and a clinical-plus-lifestyle model are trained through an identical pipeline, on identical folds, with identical random seeds and an identical hyperparameter search budget, so that any observed gain is attributable to the lifestyle modality and not to pipeline differences.

Logistic Regression, Decision Trees, Random Forests and Support Vector Machines are selected as base learners on the basis of documented performance in comparable published studies on hybrid clinical-behavioural datasets, and are benchmarked against both a reproduced conventional clinical risk score and published external results. Model transparency is delivered through the complementary, and deliberately non-redundant, application of SHapley Additive exPlanations (SHAP) for globally consistent, additively decomposed feature attribution and Local Interpretable Model-agnostic Explanations (LIME) for fast, sparse, patient-level surrogate explanations at the point of care. The resulting system is positioned strictly as a clinical decision-support tool intended to shorten the interval between the emergence of modifiable risk and its clinical recognition, and not as a replacement for professional clinical judgement.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
AUC	Area Under the Curve
BMI	Body Mass Index
BRFSS	Behavioral Risk Factor Surveillance System
CHD	Coronary Heart Disease
CVD	Cardiovascular Disease
DSR	Design Science Research
EHR	Electronic Health Record
GDPR	General Data Protection Regulation
HbA1c	Glycated Haemoglobin
KNN	K-Nearest Neighbours
LIME	Local Interpretable Model-agnostic Explanations
LR	Logistic Regression
MAR / MCAR / MNAR	Missing At Random / Missing Completely At Random / Missing Not At Random
ML	Machine Learning
NCD	Non-Communicable Disease
NHANES	National Health and Nutrition Examination Survey
NHS	National Health Service
RF	Random Forest
RFE	Recursive Feature Elimination
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
T2D	Type 2 Diabetes
UCI	University of California, Irvine (Machine Learning Repository)
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Timely detection and intervention remain the cornerstones of modern medicine. Detecting health anomalies at their inception improves survival, widens the range of viable treatments, and reduces the long-run cost of care. This is not a rhetorical claim; it is a measurable one, and the scale of the problem it describes can be stated precisely.

The World Health Organization (2025) reports that non-communicable diseases (NCDs) caused at least **43 million deaths in 2021**, representing approximately **75% of all non-pandemic-related deaths globally**. Of these, **18 million people died before the age of 70**, and 82% of those premature deaths occurred in low- and middle-income countries. Broken down by condition, cardiovascular diseases accounted for at least **19 million deaths**, cancers for 10 million, chronic respiratory diseases for 4 million and diabetes (including diabetes-attributable kidney disease) for more than 2 million. Together these four groups account for 80% of all premature NCD deaths. Critically for a study concerned with modifiable risk, the same source attributes **25% of global NCD deaths to elevated blood pressure alone** — a measurable clinical parameter that is itself strongly conditioned by diet, physical activity, alcohol intake and tobacco use.

The economic dimension of this burden is equally quantifiable, and in the United Kingdom it has been costed directly. Shih *et al.* (2025), in a cost-of-illness analysis published in the *European Heart Journal – Quality of Care and Clinical Outcomes*, estimate the total economic burden of cardiovascular disease in the UK at **£29.02 billion in 2021/22**. Of this, **£16.62 billion represents direct health and care costs**, including £6.73 billion of inpatient care, £4.65 billion of long-term care and £1.94 billion of medication, while the remaining £12.40 billion consists of informal care (£6.38 billion) and production losses from mortality and morbidity. Diabetes carries a comparable burden: Hex *et al.* (2024) estimate the total cost of diabetes to the UK at almost **£14 billion in 2021/22**, of which more than **£10 billion falls directly on the NHS**, with roughly £6 billion of that total attributable to complications rather than to the underlying diagnosis and its routine treatment.

The proportion of that burden that is attributable specifically to *late* detection is the motivating gap for this research. Three figures make the case concrete:

- **Late-stage diagnosis remains the norm for a large minority of patients.** NHS England (2025) reported that between September 2023 and August 2024, 120,958 of 206,038 staged common cancers — **58.7%** — were diagnosed at stage 1 or 2. This was the highest early-diagnosis rate ever recorded in England, and yet it still means that **41.3% of patients were not identified until stage 3 or 4**, when treatment is more invasive, more expensive and less likely to succeed.
- **A large population is undiagnosed at any given moment.** Diabetes UK (2025) estimates that almost 6 million people in the UK are living with diabetes, of whom over 4.7 million

have a formal diagnosis. That leaves approximately **1.3 million people living with undiagnosed type 2 diabetes**, alongside a further **6.9 million adults at increased risk** on the basis of blood glucose measurements — more than 12 million adults in total living with diabetes or prediabetes.

- **The cost is concentrated in the avoidable consequences of delay.** Roughly 60% of the annual NHS diabetes spend, which Diabetes UK (2025) places at £10.7 billion, is consumed by complications rather than by the condition itself. (This charity estimate and the £10 billion direct NHS figure of Hex *et al.* (2024) are independent estimates of the same quantity produced on different costing bases; both are reported here rather than reconciled.) Each week in the UK, diabetes-related complications account for approximately 184 amputations, 1,010 strokes, 700 heart attacks and 3,340 cases of heart failure.

Table 1.1: Quantified scale of the problem addressed by this research

Dimension of the problem	Quantified figure	Source
Global NCD mortality	At least 43 million deaths in 2021 (≈75% of non-pandemic deaths); 18 million premature deaths before age 70	WHO (2025)
Global CVD mortality	At least 19 million deaths in 2021	WHO (2025)
Attributable to elevated blood pressure	25% of all global NCD deaths	WHO (2025)
UK economic burden of CVD	£29.02 bn total (2021/22); £16.62 bn direct health and care costs	Shih *et al.* (2025)
UK cost of diabetes	≈£14 bn total (2021/22); >£10 bn direct NHS cost	Hex *et al.* (2024)
NHS diabetes spend on complications	≈60% of a £10.7 bn annual spend	Diabetes UK (2025)
Late-stage cancer diagnosis in England	41.3% of staged common cancers still diagnosed at stage 3 or 4 (Sep 2023 – Aug 2024)	NHS England (2025)
Undiagnosed type 2 diabetes in the UK	≈1.3 million people; a further 6.9 million at increased risk	Diabetes UK (2025)

Against this background, Machine Learning (ML) offers the capacity to process the large volumes of routinely collected patient information generated through primary-care check-ups, electronic health records and population health surveys, and to identify statistical patterns that precede clinical presentation. This project is motivated specifically by the unexploited predictive signal in *behavioural* data. Lifestyle variables — smoking history, physical activity, dietary composition, alcohol consumption — are collected routinely, are causally implicated in the pathogenesis of cardiovascular disease and type 2 diabetes, and are, unlike age or genotype, *modifiable*. A prediction that is driven in part by modifiable factors is not merely a forecast; it is an actionable intervention target. Finally, because clinicians carry legal and ethical responsibility for the decisions they make, this project is motivated to build transparency into the system from the outset through Explainable AI (XAI), rather than treating interpretability as a post-hoc addition.

1.2 Problem Statement

The problem this research addresses can be stated in four parts.

First, a substantial share of chronic disease is still identified too late. As quantified in Section 1.1, 41.3% of staged common cancers in England are diagnosed at stage 3 or 4, and approximately 1.3 million people in the UK have undiagnosed type 2 diabetes. Traditional diagnostic pathways are reactive by construction: they are typically initiated when a patient presents with symptoms, which for cardiovascular disease and type 2 diabetes may occur years after the underlying pathology becomes measurable.

Second, conventional clinical risk scores discriminate only moderately well. The established risk equations used in practice are not failing outright, but their performance leaves substantial room for improvement. Weng *et al.* (2017), analysing 378,256 UK primary-care patients, measured the ACC/AHA baseline risk algorithm at an AUC of **0.728**. Alaa *et al.* (2019), analysing 423,604 UK Biobank participants, measured the Framingham score at an AUC of **0.724**. An AUC in the low 0.7s means that a non-trivial fraction of patients who will go on to experience an event are not flagged by the instrument that is supposed to flag them.

Third, behavioural data is collected but under-exploited, and where it is used it is used crudely. Population health surveys and primary-care records routinely capture smoking status, activity levels and dietary indicators, yet most deployed risk scores incorporate only a small, fixed subset of them. Where research studies do incorporate lifestyle variables, the dominant technique is simple concatenation of lifestyle columns onto the clinical feature matrix — *early fusion* — which implicitly assumes that a self-reported, potentially noisy behavioural variable should be treated by the learning algorithm on exactly the same footing as a laboratory-measured biomarker. Section 1.4 and Chapter 2 argue that this assumption is unsafe and that it is precisely where a methodological contribution can be made.

Fourth, the models that perform best are the least transparent. The clinical adoption of machine learning is bottlenecked less by accuracy than by accountability. A clinician who cannot inspect why a model has assigned a patient to a high-risk category cannot defend acting on it, and cannot detect when the model has latched onto a spurious correlation. Ghassemi, Oakden-Rayner and Beam (2021) argue forcefully that current post-hoc explainability methods do not yet deliver the assurance that is often claimed for them, which makes the design of the explanation layer — rather than its mere presence — a substantive research question.

There is, therefore, a need for a prediction system that is early-stage in its target, principled rather than naive in how it integrates behavioural data with clinical data, rigorous in demonstrating that the behavioural data actually adds something, and transparent in a way that is useful at the point of care.

1.3 Clarification of Scope: Target Disease and Framework Generality

An early formulation of this project referred generally to "chronic and life-threatening diseases", listing cardiovascular disease, diabetes and cancer. That formulation is too diffuse to be evaluated, because the feature space, the data sources, the clinically meaningful prediction horizon and the appropriate baseline differ substantially between those conditions. This section therefore states the scope of the study precisely.

This study is a disease-specific instantiation of a disease-agnostic pipeline. Concretely:

- **Primary target disease: cardiovascular disease (CVD).** The predictive task is binary classification of the presence of, or elevated risk of, cardiovascular disease from a combined clinical and lifestyle feature vector. CVD is selected as the primary target because it carries the largest attributable mortality of any NCD group (WHO, 2025), because its risk factors are unusually well characterised and include several directly modifiable behavioural determinants, and because it provides established, published baselines — the Framingham and ACC/AHA risk equations — against which any new model can be honestly measured.
- **Secondary target disease: type 2 diabetes (T2D).** The same pipeline, unchanged in its architecture, preprocessing logic and explanation layer, is re-fitted on a type 2 diabetes dataset. The purpose of this second application is *not* to claim a general multi-disease classifier. It is a transferability test: it establishes whether the methodological contributions of this work (late fusion, interaction features, missingness-aware preprocessing) reproduce on an independent condition, or whether they are artefacts of one dataset.
- **Not in scope: a generalised multi-disease classification framework.** The system does not attempt to distinguish between diseases, does not perform differential diagnosis, and makes no claim to generalise to conditions not tested. Cancer is explicitly excluded as a modelling target, notwithstanding its use in Section 1.1 to evidence the general phenomenon of late diagnosis, because early cancer detection depends primarily on imaging and histopathology rather than on the structured tabular clinical and lifestyle features that this study is concerned with.

Table 1.2: Scope of the study — what is and is not within the target of the pipeline

Aspect	In scope	Out of scope
Target condition	Cardiovascular disease (primary); type 2 diabetes (secondary, transferability test)	Cancer, respiratory disease, differential diagnosis across conditions
Prediction task	Binary risk classification with calibrated probability output	Multi-class disease identification; staging; prognosis of time-to-event
Data modality	Structured tabular clinical measurements and self-reported lifestyle variables	Medical imaging, genomics, free-text clinical notes, waveform data
Data provenance	Publicly available, anonymised secondary datasets (UCI, Kaggle, BRFSS-derived)	Live EHR integration, wearable IoT streams, identifiable patient records
Deployment claim	Assistive clinical decision-support and risk triage	Autonomous diagnosis; replacement of clinical judgement; regulated medical device use

1.4 Research Gap and Statement of Novelty

It must be conceded at the outset that *merging tabular lifestyle and clinical data is already widely practised in health informatics*. Dinh *et al.* (2019) combined questionnaire and laboratory variables from NHANES; Xie *et al.* (2019) built type 2 diabetes risk models entirely from behavioural survey data; Alaa *et al.* (2019) fitted models over 473 UK Biobank variables spanning both modalities. "Clinical plus lifestyle plus XAI" is therefore not, by itself, a defensible novelty claim, and this report does not advance one. The contribution of this work lies in *how* the two modalities are integrated, and in the evidentiary standard applied to the claim that the integration helps. Four specific claims are made.

Novelty 1 — Dual-branch late fusion with reliability weighting, in place of early fusion by concatenation. The dominant approach in the literature is early fusion: lifestyle columns are appended to the clinical feature matrix and a single learner is fitted over the union. This treats a self-reported estimate of weekly exercise as epistemically equivalent to a laboratory-measured glucose value. This study instead trains a *clinical branch* and a *lifestyle branch* as separate base learners, calibrates each branch's probability output independently, and combines them through a stacked meta-learner. Crucially, the meta-learner receives, alongside the two branch probabilities, a per-record *lifestyle reliability score* derived from the completeness and internal consistency of that patient's behavioural responses, allowing the fusion weight to vary by record rather than being fixed globally. The clinical-branch prediction therefore degrades gracefully rather than being corrupted when behavioural data is sparse or implausible.

Novelty 2 — Explicitly engineered clinical × lifestyle interaction features. Rather than relying on tree ensembles to discover multiplicative effects implicitly, this study constructs a defined set of interaction terms that encode documented pathophysiological effect modification (for example, smoking exposure × systolic blood pressure, BMI × physical inactivity, and reported sodium/dietary quality × blood pressure). These are declared *a priori*, are individually inspectable in the SHAP output, and are subject to their own ablation, so their contribution can be reported separately from that of the raw lifestyle variables.

Novelty 3 — Missingness-aware handling of noisy behavioural data. Behavioural data is not missing at random: patients who decline to report alcohol intake or activity level differ systematically from those who report it. Weng *et al.* (2017) observed exactly this, finding that the *absence* of a recorded BMI value was itself an informative predictor in their neural network. This study therefore retains binary missingness-indicator features rather than discarding them through imputation, and treats the imputation strategy itself (mode versus KNN) as an experimental variable evaluated on both synthetic and real-world data, following the evaluation design of Beretta and Santaniello (2016).

Novelty 4 — A strict, pipeline-controlled ablation as the primary test of the claim. The central empirical claim of this research is tested by a single, tightly controlled comparison: a *clinical-only* model and a *clinical-plus-lifestyle* model trained through the identical pipeline, on identical cross-validation folds, with identical random seeds, identical hyperparameter search budgets, identical class-balancing and identical calibration. Any difference in performance is therefore attributable to the lifestyle modality rather than to incidental pipeline variation, and is assessed for statistical significance rather than reported as a raw difference in point estimates. The full benchmark design is specified in Section 3.8.

Taken together, these four claims convert a general and already-crowded proposition ("integrate lifestyle data and explain the model") into a set of specific, falsifiable design decisions, each of which can be shown by experiment to help or not to help.

1.5 Research Questions

- **RQ1.** What is the measurable discriminative performance of a conventional clinical risk-scoring approach when reproduced on this study's cohort, and how does it compare with a clinical-only machine learning model trained on the same data? *(Answered by benchmark configurations B0 and B1 in Section 3.8. Note that this question is deliberately confined to discrimination; the datasets available to this study are cross-sectional, as Section 1.8 records, so time-to-detection cannot be measured and is not claimed.)*
- **RQ2.** Does the addition of lifestyle and behavioural variables to a clinical feature set produce a statistically significant improvement in predictive performance when the pipeline is otherwise held constant, and how large is that improvement?
- **RQ3.** Does a dual-branch late-fusion architecture with reliability weighting outperform conventional early fusion by concatenation on the same data, and does the addition of explicit clinical × lifestyle interaction features add further measurable value?
- **RQ4.** How does the choice of imputation strategy for missing self-reported behavioural data (mode imputation versus KNN imputation, with and without retained missingness indicators) affect downstream predictive performance, and does that effect differ between synthetically generated missingness and real-world clinical missingness?
- **RQ5.** How do Logistic Regression, Decision Trees, Random Forests and Support Vector Machines compare on the hybrid dataset, and how do the resulting models compare against a reproduced conventional clinical risk score and against published external benchmarks?
- **RQ6.** What distinct and non-redundant information do SHAP and LIME each contribute to clinical interpretation, and to what extent do their attributions agree at the level of the individual patient?

1.6 Research Objectives

1. To reproduce a conventional clinical risk score on the study cohort and to establish a clinical-only machine learning baseline against it, providing the performance floor for all subsequent comparisons (RQ1).
2. To assemble and preprocess hybrid clinical-and-lifestyle datasets for a primary cardiovascular target and a secondary type 2 diabetes target, documenting missingness mechanisms and class balance for each.
3. To implement a reproducible preprocessing pipeline incorporating missingness indicators, a controlled comparison of mode and KNN imputation, scaling appropriate to each learner, and leakage-safe cross-validation.
4. To engineer and document a declared set of clinical × lifestyle interaction features grounded in the pathophysiological literature.
5. To develop, tune and evaluate Logistic Regression, Decision Tree, Random Forest and Support Vector Machine models, both as single learners over concatenated features and as branches within the proposed late-fusion architecture.

6. To execute the ablation benchmark specified in Section 3.8, comparing clinical-only against clinical-plus-lifestyle models under identical conditions, and to test the resulting difference in AUC for statistical significance.
7. To evaluate all models using Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC and calibration, with Recall and calibration treated as primary given the clinical cost asymmetry of a false negative.
8. To integrate SHAP for global and additively consistent attribution and LIME for local, patient-level surrogate explanation, and to quantify the agreement between the two at the individual-prediction level.
9. To document the ethical, data-governance and fairness position of the system, including subgroup performance analysis across available demographic strata.

1.7 Research Aims

1. **Build a hybrid predictive system whose integration strategy is principled and tested.** To construct a late-fusion machine learning system for early-stage cardiovascular disease risk in which the contribution of the lifestyle modality is demonstrated by controlled experiment rather than asserted.
2. **Deliver transparency that is actually usable at the point of care.** To integrate SHAP and LIME in complementary, clearly differentiated roles — global model audit and per-patient explanation respectively — and to report honestly on where their explanations diverge.
3. **Produce a decision-support tool oriented toward earlier risk recognition.** To provide clinicians with calibrated, explained risk estimates that identify modifiable behavioural contributors to an individual patient's risk, supporting better-targeted preventative intervention. Because the available data is cross-sectional, this aim is pursued through improved discrimination and interpretability on risk that is measurable but not yet clinically recognised; a demonstrated reduction in time-to-diagnosis would require prospective data and is identified as future work.

1.8 Scope and Limitations

The scope of the target condition and prediction task is set out in Section 1.3 and Table 1.2. The following limitations are additionally acknowledged.

Secondary data. The study relies on publicly available, anonymised secondary datasets. These datasets are cross-sectional rather than longitudinal, which means the models learn association with prevalent or recorded disease status rather than incidence over a defined follow-up window. This is a real constraint on the strength of any "early prediction" claim, and it is stated explicitly in the interpretation of results rather than glossed over. Where a dataset is known or suspected to be partly synthetic or synthetically balanced, this is recorded in Table 3.1 and its results are reported separately from results on wholly observational data.

Self-report bias. Lifestyle variables in all of the selected datasets are self-reported and are therefore subject to recall error and social-desirability bias, particularly for alcohol intake, smoking and physical activity. The reliability-weighting mechanism described in Section 3.6 mitigates but does not eliminate this.

No real-time integration. Live EHR integration and continuous wearable data streams are outside the scope of this prototype and are identified as future work.

Not a medical device. The system is an assistive decision-support prototype developed for academic assessment. It has not undergone clinical validation, regulatory assessment or prospective evaluation, and it is not intended to replace clinical examination or professional judgement.

1.9 Thesis Structure

- **Chapter 1: Introduction** — quantifies the clinical and economic scale of the problem, fixes the scope of the target disease, and states the specific novelty claims and research questions of the study.
- **Chapter 2: Literature Review** — reviews prior empirical studies that applied the selected algorithms to hybrid clinical-behavioural datasets, reports the performance those studies achieved, and derives from that evidence both the selection of baseline algorithms and the specific methodological gap this project occupies.
- **Chapter 3: Methodology** — specifies the datasets, preprocessing protocol, fusion architecture, benchmark and ablation design, evaluation metrics and XAI integration.
- **Chapter 4: Results and Analysis** — will present the empirical performance of each configuration and the outputs of the explanation layer. *(To be completed in the subsequent phase.)*
- **Chapter 5: Discussion** — will interpret the findings against the research questions and the published benchmarks identified in Chapter 2. *(To be completed in the subsequent phase.)*
- **Chapter 6: Conclusion and Future Work** — will summarise outcomes, state limitations and identify extensions. *(To be completed in the subsequent phase.)*

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction and Review Strategy

The purpose of this chapter is not to define machine learning algorithms. Definitions of Logistic Regression, Decision Trees, Random Forests and Support Vector Machines are available in any standard text (Bishop, 2006; Kuhn and Johnson, 2013; James *et al.*, 2021; Géron, 2022) and are assumed. The purpose of this chapter is instead **methodological**: to review what previous researchers actually achieved when they applied these specific algorithms to hybrid clinical-and-behavioural healthcare data, to use those reported results to justify the algorithm selection made in Chapter 3, and to identify the precise methodological gap that this project occupies.

Literature was identified through structured searches of PubMed, IEEE Xplore, ScienceDirect, SpringerLink and Google Scholar using combinations of the terms *cardiovascular risk prediction*, *type 2 diabetes prediction*, *machine learning*, *lifestyle factors*, *behavioural risk factors*, *data fusion*, *missing data imputation*, *SHAP* and *LIME*. Priority was given to studies that (a) reported quantitative discriminative performance, (b) used cohorts or datasets containing both clinical and behavioural variables, and (c) were published in peer-reviewed venues. Textbook sources are retained only where a definitional or theoretical anchor is required.

The chapter proceeds from the clinical case for early prediction (2.2), through the measured limitations of conventional risk scores (2.3), to the empirical evidence base for the four selected algorithms (2.4) and for the incremental value of lifestyle data (2.5). Section 2.6 reviews the specific technical literature on imputing missing and noisy behavioural data. Section 2.7 addresses explainability, and in particular the question of why SHAP and LIME are both required rather than one being chosen over the other. Sections 2.8 and 2.9 cover ethics and synthesise the research gap.

Note on numbering: this chapter has been restructured since the first review and its section numbers have therefore changed. The comparison of SHAP and LIME, which was previously Section 2.6.2, is now Section 2.7, and the specific argument for why both methods are required is Section 2.7.4. Section 2.6 in this version is the new review of imputation literature.

2.2 The Clinical and Economic Case for Early Prediction

Early prediction is valuable in proportion to the size of the gap between the point at which risk becomes measurable and the point at which disease becomes clinically apparent. For cardiovascular disease and type 2 diabetes that gap is large. The World Health Organization (2025) attributes 25% of global NCD deaths to elevated blood pressure, a parameter that is routinely measured, cheaply monitored and responsive to behavioural modification years before an acute event. Diabetes UK (2025) estimates 1.3 million people in the UK are living with undiagnosed type 2 diabetes and a further 6.9 million with elevated blood glucose short of a diagnostic threshold — a population that is, by definition, measurable but unidentified.

The economic argument follows from the clinical one. Because approximately 60% of the annual NHS diabetes spend is consumed by complications (Diabetes UK, 2025), and because £6.73 billion of the UK's £16.62 billion direct CVD care costs is inpatient care (Shih *et al.*, 2025), interventions that move detection earlier in the disease trajectory act on the largest and most avoidable components of expenditure. This is what makes a decision-support tool aimed at earlier risk recognition economically as well as clinically consequential.

2.3 Measured Limitations of Traditional Diagnosis and Conventional Risk Scores

2.3.1 The reactive character of symptomatic diagnosis

Traditional diagnostic workflows are triggered by symptomatic presentation. For conditions with a long asymptomatic phase this means the diagnostic process starts late by construction. The NHS England (2025) staging data quantifies the consequence in the analogous case of cancer: even at the highest early-diagnosis rate ever recorded in England, 41.3% of staged common cancers were identified only at stage 3 or 4.

2.3.2 The discriminative ceiling of conventional risk equations

Established cardiovascular risk equations perform moderately, not well. Two large UK studies quantify this precisely and provide the external benchmarks used in Chapter 3.

Weng *et al.* (2017) applied four machine learning algorithms to routine primary-care data from 378,256 patients (295,267 training, 82,989 validation) and compared them against the established ACC/AHA risk algorithm. The ACC/AHA baseline achieved an AUC of **0.728**. All four machine learning approaches exceeded it: Random Forest 0.745, Logistic Regression 0.760, Gradient Boosting Machines 0.761 and Neural Networks 0.764. The improvement is modest in absolute terms but consequential at population scale. Notably for the present study, the machine learning models selected predictors that conventional equations exclude, including chronic obstructive pulmonary disease, severe mental illness, oral corticosteroid prescribing, triglycerides and HbA1c — and, strikingly, *the absence of a recorded BMI value*, which the neural network treated as an informative signal in its own right.

Alaa *et al.* (2019) reached a compatible conclusion on a different cohort. Across 423,604 UK Biobank participants with a median follow-up of seven years and 4,801 cardiovascular events within the five-year evaluation window, the Framingham score achieved an AUC-ROC of **0.724** (95% CI 0.720–0.728), while an automated machine learning pipeline fitted over 473 available variables achieved **0.774** (95% CI 0.768–0.780). The variables that drove the improvement are directly relevant here: among the most informative predictors identified were *usual walking pace* and *self-reported overall health rating* — behavioural and self-reported variables that, as the authors observe, "are not usually considered in existing risk prediction models".

These two studies establish the empirical premise of this project. Conventional clinical risk scores sit at an AUC of roughly 0.72–0.73; machine learning over a richer feature set, including behavioural variables, reaches roughly 0.76–0.79 across the studies reviewed in Section 2.4. The

remaining question — the one this project addresses — is whether the integration strategy for those behavioural variables can be improved beyond naive inclusion.

2.4 Prior Studies Applying the Selected Algorithms to Hybrid Healthcare Datasets

This section reviews the performance that previous researchers obtained using Logistic Regression, Decision Trees, Random Forests and Support Vector Machines on datasets combining clinical and behavioural variables, and derives from that evidence the justification for selecting these four as the baseline algorithms of this study.

Table 2.1: Prior studies applying the selected baseline algorithms to hybrid clinical-lifestyle datasets

Study	Data and target	Feature modalities	Reported performance	Relevance to this study
Weng *et al.* (2017)	378,256 UK primary-care patients; 10-year CVD	Routine clinical records including smoking, BMI, deprivation	AUC: ACC/AHA baseline 0.728; RF 0.745; LR 0.760; GBM 0.761; NN 0.764	External CVD benchmark; evidences informative missingness
Alaa *et al.* (2019)	423,604 UK Biobank participants; 5-year CVD	473 variables: clinical, biomarker, lifestyle, self-reported	AUC-ROC: Framingham 0.724; automated ML 0.774	Quantifies the ceiling reachable by adding behavioural variables
Dinh *et al.* (2019)	NHANES 1999–2014 / 2007–2014; CVD and diabetes	Questionnaire (lifestyle) vs. questionnaire + laboratory	CVD AUC 0.831 without labs, 0.839 with labs; diabetes AUC 0.862 without labs, 0.957 with labs. LR, SVM, RF, XGBoost and a weighted ensemble were compared, but per-algorithm figures were not separately reported	The closest existing analogue to the ablation proposed here
Xie *et al.* (2019)	2014 BRFSS, 138,146 participants (20,467 with T2D)	Predominantly behavioural: sleep, physical activity, exercise frequency, smoking, alcohol, general health	AUC / accuracy: NN 0.7949 / 82.4%; LR 0.7932 / 80.7%; linear SVM 0.7807 / 80.8%; RBF SVM 0.7788 / 81.8%; RF 0.7608 / 79.3%; decision tree 0.7182 / 74.3%	Direct head-to-head evidence on LR, DT, RF and SVM on behavioural data
Mohan, Thirumalai and Srivastava (2019)	UCI Cleveland heart disease subset	Clinical attributes	Hybrid random forest with linear model (HRFLM) reported at 88.7% accuracy, exceeding the individual constituent learners	Precedent for hybridising a linear and an ensemble learner
Islam *et al.* (2020)	UCI Early Stage Diabetes Risk Prediction dataset (520 records)	Symptom and behavioural indicators, no laboratory values	Of Naive Bayes, Logistic Regression and Random Forest, Random Forest was reported as achieving the best accuracy under tenfold cross-validation	Secondary-target dataset precedent; shows small-sample optimism risk

Study	Data and target	Feature modalities	Reported performance	Relevance to this study
Paul, Ibrahim and Masood (2026)	≈70,000 records combining the Cardio dataset and UCI sources	Clinical plus lifestyle (smoking, alcohol, activity)	AUC: Gradient Boosting 0.794; voting ensemble 0.793; RF 0.792; LR 0.773. Accuracy ≈0.72–0.73, F1 0.70–0.72	Directly comparable primary-dataset benchmark; SHAP-only explanation layer

2.4.1 What this evidence justifies about the selection of baseline algorithms

Four conclusions follow from Table 2.1, and each maps onto a design decision in Chapter 3.

Logistic Regression is retained as the baseline because it is empirically competitive, not merely because it is simple. This is the most important and most counter-intuitive finding in the evidence base. Weng *et al.* (2017) found Logistic Regression (AUC 0.760) outperformed Random Forest (0.745) on large-scale primary-care data. Xie *et al.* (2019) found Logistic Regression second only to a neural network (AUC 0.7932 against 0.7949), ahead of both SVM variants and well ahead of Random Forest (0.7608). A well-specified linear model on tabular clinical data is a demanding benchmark, not a straw man, and a hybrid architecture that cannot beat it has not demonstrated anything.

Support Vector Machines are retained because they are competitive but kernel-sensitive. Xie *et al.* (2019) reported a linear SVM at AUC 0.7807 and an RBF SVM at 0.7788 — close to each other in AUC but differing in accuracy (80.8% versus 81.8%), which indicates sensitivity to the decision threshold and to kernel choice. This justifies including both kernels in the hyperparameter search of Section 3.7 rather than committing to one.

Random Forests are retained because they capture interactions that linear models cannot, while not dominating uniformly. Random Forest was outperformed by Logistic Regression in both Weng *et al.* (2017) and Xie *et al.* (2019), yet performed strongly in Paul, Ibrahim and Masood (2026) at AUC 0.792 and in Islam *et al.* (2020). The inconsistency is informative: ensembles gain where feature interactions are strong and the sample is large enough to estimate them. Because this study explicitly hypothesises that clinical × lifestyle *interactions* carry signal (Novelty 2), a learner capable of representing interactions natively is required as a comparator against the explicitly engineered interaction terms.

Decision Trees are retained as an interpretability reference point, with realistic expectations. The single decision tree was the weakest classifier in Xie *et al.* (2019), at AUC 0.7182 and 74.3% accuracy — below every other model tested, and in fact *below* the ACC/AHA clinical baseline of 0.728 reported by Weng *et al.* (2017) on a different cohort. A single tree is therefore not a credible predictive candidate in its own right. It is retained because it provides a directly human-readable rule set against which the SHAP and LIME explanations of the stronger models can be sanity-checked, and because the gap between it and the Random Forest quantifies precisely how much performance is being traded for intrinsic interpretability — a trade-off that is central to the argument of Section 2.7.

A further methodological caution is drawn from Table 2.1. Islam *et al.* (2020) report a headline accuracy on a 520-record dataset that appears far higher than the AUC 0.764 that Weng *et al.* (2017) obtained on 378,256 patients. That apparent superiority is an artefact of sample size and of the comparative ease of the smaller study's prediction task, not evidence of a better model. This observation directly motivates the requirement in Section 3.8 that comparisons be made *within* a controlled pipeline on identical folds, and that headline accuracy figures from small datasets never be used as evidence of methodological progress.

2.5 Evidence on the Incremental Value of Lifestyle Data

The premise that behavioural data adds predictive value is testable, and several studies have effectively tested it. Table 2.2 summarises the evidence.

Table 2.2: Evidence on the incremental predictive value of lifestyle and behavioural variables

Source	Finding	Implication for this study
Dinh *et al.* (2019)	For cardiovascular disease, models built *without* laboratory results reached AUC 0.831; adding laboratory results raised this only to 0.839. For diabetes the corresponding figures were 0.862 and 0.957.	Behavioural and questionnaire data alone carry most of the CVD signal, but not the diabetes signal. The two modalities are not interchangeable, and their relative weight is disease-dependent — a direct argument for late fusion with learned weights rather than fixed concatenation.
Dinh *et al.* (2019), feature ranking	Top diabetes predictors included waist size, age, self-reported weight, leg length and *sodium intake*; top CVD predictors included age, systolic and diastolic blood pressure, self-reported weight and chest pain.	Dietary sodium ranking in the top five for diabetes supports the specific interaction terms proposed in Section 3.5.
Alaa *et al.* (2019)	Usual walking pace and self-reported overall health rating were among the most informative of 473 variables, and are absent from conventional risk equations.	Confirms that the marginal value of behavioural data lies in variables that clinical scores currently discard.
Xie *et al.* (2019)	Models built almost entirely from behavioural survey variables achieved AUC up to 0.7949 (accuracy 82.4%) for type 2 diabetes across 138,146 participants.	Establishes that a lifestyle-only branch is viable as an independent base learner — the precondition for the dual-branch architecture proposed in Section 3.6.
Weng *et al.* (2017)	The *absence* of a recorded BMI value functioned as an informative predictor in the neural network model.	Missingness in behavioural and anthropometric data is itself signal, and should be retained rather than imputed away.

2.5.1 Fusion strategy: what the multimodal literature establishes

Because the central methodological claim of this project concerns *how* two modalities are combined, the literature on fusion strategy itself must be reviewed rather than assumed. That literature is well developed outside the specific case of tabular clinical and lifestyle data, and it supplies both a taxonomy and a warning.

Stahlschmidt, Ulfenborg and Synnergren (2022) review multimodal fusion for biomedical data and distinguish three strategies. In **early fusion**, features from all modalities are concatenated at the input and a single model is fitted over the union; the approach is simple but, in the authors'

assessment, "may not identify relationships between modalities when they only become apparent at higher levels of abstraction". In **intermediate (joint) fusion**, a representation is learned separately for each modality and the representations are combined at a chosen depth. In **late fusion**, separate models are trained per modality and their predictions are aggregated; this accommodates heterogeneous data types well, but the authors are explicit that it "cannot learn interactions between features of different modalities".

Huang *et al.* (2020) provide the corresponding empirical picture in a systematic review of 17 studies fusing medical imaging with electronic health record data, screened from 985 candidates. Of these, 11 used early fusion, four joint fusion and three late fusion. Their headline finding is that "multimodality fusion models generally led to increased accuracy (1.2–27.7%) and AUROC (0.02–0.16) over traditional single modality models", but also — and this is the finding that shapes the design of this study — that "no single fusion strategy consistently led optimal performance across all domains". All three late-fusion studies improved on single-modality baselines, as did six of seven comparative early-fusion studies.

Two consequences follow for the present work, and the second is a direct constraint on the design.

First, because no fusion strategy dominates across domains, the choice between early and late fusion for tabular clinical and lifestyle data is an *empirical question for this data*, not a matter of adopting the best-performing method from another domain. This is precisely why Section 3.8 specifies configuration B2 (early fusion) and configuration B3 (late fusion) as a head-to-head comparison on identical folds, rather than presenting late fusion as an assumed improvement.

Second, the criticism that late fusion cannot represent cross-modal interactions (Stahlschmidt, Ulfenborg and Synnergren, 2022) is a genuine and serious objection to the architecture proposed in Section 3.6, and it is not evaded here. It is, however, the specific reason the two design decisions of this project are paired rather than independent: the declared clinical × lifestyle interaction features of Section 3.5.2 are computed *before* the branch split and supplied to the clinical branch, so that the cross-modal interaction terms the late-fusion structure would otherwise discard are reintroduced explicitly and in an auditable form. Late fusion is chosen for its ability to weight a noisy self-reported modality differently from a measured one; explicit interaction features are the compensation for what late fusion gives up in exchange. Configuration B4 in Table 3.3 tests exactly this compensation, and if it fails to recover the lost interaction signal, that is a reportable negative result about the architecture rather than an inconvenience to be omitted.

2.5.2 The gap this evidence exposes

Every study in Table 2.2 integrates the two modalities by early fusion, placing all variables into a single feature matrix. None of them models the two modalities separately and learns how much to trust each. Late fusion by stacked generalisation is not itself a novel technique — it is long established (Wolpert, 1992) and is one of the three standard strategies catalogued by Stahlschmidt, Ulfenborg and Synnergren (2022) — and no claim of novelty is made for it here. What has not been done, on the evidence reviewed, is its application to the specific problem of

combining measured clinical variables with self-reported behavioural variables in tabular risk prediction, with the fusion weight conditioned on the reliability of the behavioural record. Yet Dinh *et al.* (2019) provide the clearest possible motivation for doing so: the marginal value of laboratory data over questionnaire data was 0.008 of AUC for cardiovascular disease but 0.095 for diabetes — an order of magnitude difference between the two conditions. A fusion mechanism that can learn this difference, rather than having it imposed by the feature matrix, is the specific methodological opportunity this project pursues.

2.6 Preprocessing Literature: Missing and Noisy Behavioural Data

(This section is new. Readers looking for the SHAP and LIME discussion that was numbered 2.6.2 in the previous version will find it at Section 2.7 below, with the comparison of the two methods at Section 2.7.4.)

Because self-reported lifestyle variables are the noisiest and most frequently missing part of the feature space, the treatment of missing data is not a routine preprocessing footnote in this project but a substantive methodological choice with published evidence behind it. This section reviews that evidence, since the methodology of Chapter 3 depends on it.

2.6.1 Machine learning imputation versus statistical imputation

Jerez *et al.* (2010) conducted one of the most directly relevant evaluations, comparing statistical imputation methods (mean/mode substitution, hot-deck imputation and multiple imputation) against machine learning methods (multi-layer perceptron, self-organising maps and K-Nearest Neighbours) on a real breast cancer prognosis dataset. They reported that the machine learning imputation approaches produced statistically significant improvements in downstream prognostic classification relative to the simpler statistical procedures. Their result is the primary published justification for preferring KNN over mode imputation on real-world clinical data, and it is the reason KNN imputation is included in this project's design.

2.6.2 The synthetic-versus-real-world distinction, and why it matters here

The more important, and more cautionary, finding comes from Beretta and Santaniello (2016), who evaluated nearest-neighbour imputation across both synthetic datasets and a real-world clinical dataset (the SPECTF heart dataset). Their design is directly analogous to the situation faced in this project. They compared 1NN, KNN with $k = 5$ and weighted KNN with $k = 5$ across five analytical frameworks, using synthetic datasets with 15% and 30% missing cases and the real SPECTF heart data with 15% missingness introduced under a missing-completely-at-random mechanism.

Their conclusion is that reported imputation accuracy can be actively misleading. Increasing k improved imputation error but *distorted the underlying data distribution*; as the authors state, "1NN was however the only method capable of preserving the data structure and data were distorted even when small values of k neighbors were considered". Their recommendation is that three neighbours in conjunction with ReliefF-based attribute filtering "seems to provide the best trade-off between imputation error and preservation of the data structure".

Table 2.3: Prior evaluations of imputation strategy on synthetic and real-world clinical data

Study	Design	Finding adopted in this project
Jerez *et al.* (2010)	Mean/mode, hot-deck and multiple imputation compared against MLP, self-organising map and KNN imputation on a real breast cancer prognosis dataset	Machine learning imputation, including KNN, yielded statistically significant improvements in downstream classification over simple statistical substitution — justifying KNN as a candidate strategy
Beretta and Santaniello (2016)	1NN, KNN (k=5) and weighted KNN evaluated on synthetic data at 15% and 30% missingness and on the real SPECTF heart dataset at 15% MCAR missingness	Larger k lowers imputation error but distorts the data distribution; a small neighbourhood (≈3) with ReliefF filtering gives the best trade-off. Imputation must therefore be validated on real clinical data, not only on synthetic benchmarks, and evaluated on distributional fidelity as well as error
Weng *et al.* (2017)	Machine learning over routine primary-care records with substantial real-world missingness	Missingness itself was predictive (absent BMI acted as a signal), so missingness indicators are retained as features rather than being erased by imputation

Three methodological commitments follow directly, and are implemented in Section 3.4. First, imputation strategy is treated as an experimental variable rather than a fixed assumption: mode imputation and KNN imputation are both evaluated, with the neighbourhood size tuned rather than defaulted. Second, imputation is evaluated on *both* synthetically introduced missingness and the natural missingness present in the real datasets, because Beretta and Santaniello (2016) demonstrate these can behave differently. Third, following Weng *et al.* (2017), binary missingness indicators are retained as model features so that informative non-response is preserved rather than destroyed.

2.7 Explainable AI for Clinical Trust

2.7.1 The requirement for transparency

The clinician who acts on a model's output carries the legal and ethical responsibility for the consequences. This asymmetry — the model produces the recommendation, the clinician absorbs the liability — is the fundamental reason that predictive accuracy alone has proved insufficient for clinical adoption. Explainable AI addresses this by exposing how input variables influence output predictions (Molnar, 2022).

2.7.2 SHAP: globally consistent, additively decomposed attribution

SHAP (Lundberg and Lee, 2017) grounds feature attribution in cooperative game theory, treating each feature as a player and each prediction as a payout to be distributed among them. Shapley values are the unique attribution satisfying local accuracy, missingness and consistency, which gives SHAP a property no heuristic importance measure possesses: attributions are *additive* and therefore comparable and aggregatable. Lundberg *et al.* (2020) extended this to tree ensembles with an exact polynomial-time algorithm and demonstrated that many local explanations can be composed into a coherent global picture of model behaviour. In practice this means SHAP can answer the questions a model *audit* asks: which factors drive risk across the

whole population, in what direction, over what range, and with what interactions.

2.7.3 LIME: local surrogate approximation

LIME (Ribeiro, Singh and Guestrin, 2016) takes a fundamentally different route. It makes no attempt to explain the global model. It perturbs the neighbourhood of a single instance, observes the black-box model's responses, and fits a simple, sparse, interpretable surrogate — typically a sparse linear model — that is faithful *locally*. The output is a short list of the handful of features that mattered for *this* prediction, in a form that requires no familiarity with the underlying model.

2.7.4 Why both are necessary, and not redundant

The supervisory challenge to this project was precise: if SHAP already produces both global summaries and per-patient waterfall plots, what does LIME add? The answer is that SHAP and LIME differ in four respects that matter specifically in a live decision-support setting, and that the differences make them complementary rather than substitutable.

Explanandum. The two methods answer formally different questions. A SHAP value is defined relative to a *background distribution*: it states how much a feature moved this patient's risk away from the expected prediction over a reference population, and the values sum exactly to the difference between this prediction and that baseline. LIME makes no reference to a population baseline at all. It fits a sparse weighted linear surrogate to the model's behaviour in a small perturbed neighbourhood of this one patient, and its coefficients describe the model's *local sensitivity*: how the prediction responds to small changes in each feature *for patients like this one*. So SHAP answers "why is this patient higher risk than the average patient?", which is a question about population positioning, while LIME answers "which features is the model actually responding to in this patient's immediate vicinity, and in which direction?", which is a question about local behaviour. The second is the more useful of the two when the clinical purpose is to identify which of a patient's modifiable factors the model is currently sensitive to. It should be stated precisely that LIME does *not* compute the minimal change required to flip a classification; that is counterfactual explanation, a distinct method family (for example Wachter, Mittelstadt and Russell, 2018) which this study identifies as an extension for future work rather than claiming as a present capability. LIME's local gradient-like weights nonetheless indicate *where* the model is most responsive, which is the necessary first step toward such a conversation.

Computational cost at the point of care. TreeSHAP's efficient exact algorithm (Lundberg *et al.*, 2020) applies only to tree-based models. In a system that deliberately includes a Support Vector Machine branch and a stacked meta-learner, SHAP for those components falls back to KernelSHAP, which requires a large number of perturbed model evaluations per instance to achieve a stable attribution. It should be acknowledged that KernelSHAP and LIME are closely related — KernelSHAP is a local surrogate fitted with a particular kernel and loss chosen so that its coefficients recover Shapley values — so the difference here is one of sample budget and stability requirement rather than of algorithmic class. That difference is nonetheless

operationally real: a LIME explanation is typically usable at a sample budget an order of magnitude smaller than a stable KernelSHAP attribution requires, which matters when the explanation must appear during a consultation rather than in an overnight batch report.

Form of the output and its audience. SHAP output is dense and quantitative: every feature receives a signed contribution, and the values sum exactly to the prediction. This is what an auditor, a regulator or a data scientist validating the model needs. LIME output is sparse and rule-like: three to five features with weights, deliberately discarding the rest. A sparse explanation is not a worse explanation; for a clinician making a decision under time pressure it is the appropriate one, and it aligns with how clinical reasoning is documented.

Cross-method disagreement as a diagnostic signal. The most substantive reason to run both is that their disagreement is informative. Because SHAP and LIME rest on different mathematical foundations — axiomatic game-theoretic attribution against local surrogate fitting — agreement between them constitutes weak convergent evidence that an explanation reflects a genuine model dependency rather than an artefact of one method's assumptions. Comparative evaluations of local interpretability techniques in healthcare (El Shawi *et al.*, 2021) assess exactly these properties of stability and consistency across methods. This project therefore treats explanation agreement as a measurable quantity in its own right (Section 3.10.3), rather than assuming either method is correct.

Table 2.4: Complementary properties of SHAP and LIME in a live decision-support setting

Property	SHAP	LIME
Theoretical basis	Cooperative game theory; Shapley values (Lundberg and Lee, 2017)	Local surrogate model fitted on perturbed neighbourhood (Ribeiro, Singh and Guestrin, 2016)
Question answered	Why is this patient's risk higher or lower than the population baseline?	Which features is the model locally most sensitive to for this patient, and in which direction?
Scope	Local attributions that aggregate consistently into a global model audit (Lundberg *et al.*, 2020)	Strictly local; no coherent global aggregation
Output form	Dense, signed, additive contributions across all features	Sparse rule-like list of a few decisive features
Computational cost	Exact and fast for tree models; expensive model-agnostic estimation for SVM and stacked models	Low and roughly constant per instance; suited to interactive use
Primary role in this project	Model audit, global feature hierarchy, interaction analysis, fairness inspection, validation of engineered features	Per-patient consultation-time explanation and identification of proximate modifiable factors

2.7.5 A necessary critical perspective

An honest review must record that the value of post-hoc explanation is contested. Ghassemi, Oakden-Rayner and Beam (2021) argue that current post-hoc explainability methods, including saliency approaches and feature-attribution methods of the SHAP and LIME family, provide a weaker guarantee than clinicians are typically led to believe: an explanation that is plausible is not thereby a correct account of the model's reasoning, and a persuasive explanation may

increase inappropriate trust rather than calibrate it. This project accepts that criticism rather than dismissing it, and responds to it in two concrete ways: by measuring rather than assuming the agreement between two independent explanation methods (Section 3.10.3), and by retaining an intrinsically interpretable model — the Decision Tree — as a reference against which the post-hoc explanations of the stronger models can be checked. The claim made by this project is therefore that explanations are a tool for scrutiny, not a warrant of correctness.

2.8 Ethical Considerations and Data Governance

2.8.1 Patient data privacy and GDPR

Health data is special category data under the UK GDPR and the Data Protection Act 2018, and its processing requires an identified lawful basis and appropriate safeguards. This project uses only publicly available, previously anonymised secondary datasets, which removes the requirement for individual consent while not removing the obligation to handle the data securely, to avoid any attempt at re-identification, and to comply with each dataset's licence terms.

2.8.2 Bias and fairness

Models inherit the biases of their training data. This risk is elevated for behavioural variables specifically, because self-reporting behaviour varies systematically with education, culture and socio-economic position, and because survey non-response is not random across demographic groups. A model that learns that non-reporting predicts risk may in effect be learning a proxy for deprivation. Because Section 3.4 deliberately retains missingness indicators as features, this project has a specific obligation to test for it, and Section 3.11 accordingly specifies subgroup performance reporting across available demographic strata rather than reporting aggregate performance alone.

2.9 Synthesis: Research Gap Identification

The literature reviewed establishes that hybrid clinical-behavioural modelling is well established and moderately effective; that conventional risk equations sit at roughly AUC 0.72–0.73 and machine learning over richer features reaches roughly 0.76–0.79; that the relative value of the two modalities varies by disease; that imputation strategy materially affects results and behaves differently on synthetic and real data; and that explanation methods are useful but contested. Table 2.5 states the resulting gap.

Table 2.5: Research gap identification matrix

Dimension	Prevailing practice in the reviewed literature	Gap	Response in this project
Integration strategy	Early fusion: lifestyle columns concatenated onto the clinical matrix and a single learner fitted (Dinh *et al.*, 2019; Alaa *et al.*, 2019; Xie *et al.*, 2019)	Self-reported behavioural variables are treated as epistemically equivalent to measured biomarkers, with a fusion weight fixed implicitly and globally	Dual-branch late fusion with per-record reliability weighting (Section 3.6)
Feature interaction	Interactions left to be discovered implicitly by ensembles	Known pathophysiological effect modification is neither declared nor individually auditable	Declared clinical × lifestyle interaction features with their own ablation (Section 3.5)
Missing behavioural data	Imputation applied as a fixed preprocessing assumption; missingness discarded	Beretta and Santaniello (2016) show imputation choice distorts distributions; Weng *et al.* (2017) show missingness is itself predictive	Imputation as an experimental variable, evaluated on synthetic and real missingness, with missingness indicators retained (Section 3.4)
Evidence for the integration claim	Improvement asserted by comparing against published results from different pipelines and cohorts	Confounds the value of the data modality with incidental pipeline differences	Pipeline-controlled ablation on identical folds and seeds, with DeLong significance testing (Section 3.8)
Explainability	A single method applied post-hoc, most commonly SHAP alone (for example Paul, Ibrahim and Masood, 2026)	No cross-method validation; explanation correctness assumed rather than tested (Ghassemi, Oakden-Rayner and Beam, 2021)	SHAP and LIME in differentiated roles, with measured agreement and an intrinsically interpretable reference model (Section 3.10)

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter specifies how the claims made in Section 1.4 will be constructed and tested. It follows directly from Chapter 2: each methodological choice below is traceable to a finding in the reviewed literature rather than adopted by convention. The chapter proceeds through research design (3.2), data sources (3.3), preprocessing (3.4), feature engineering (3.5), the fusion architecture (3.6), model development (3.7), the benchmark and ablation design (3.8), evaluation metrics (3.9), the explainability layer (3.10), ethics and governance (3.11), and implementation and timeline (3.12–3.13).

3.2 Research Design

The study adopts a quantitative, experimental design within a Design Science Research (DSR) frame, since the object of the research is the construction and evaluation of an artefact — a predictive pipeline — rather than the observation of a naturally occurring phenomenon. The experimental component is a controlled comparison of pipeline configurations, and it is this component that carries the evidentiary burden of the research questions.

The design proceeds in five stages:

1. **Baseline establishment.** A conventional clinical risk score is reproduced on the study cohort, and a clinical-only machine learning model is fitted. These provide the floor against which every subsequent configuration is measured.
2. **Naive integration.** Lifestyle variables are added by simple concatenation (early fusion), reproducing the prevailing practice identified in Chapter 2.
3. **Proposed integration.** The dual-branch late-fusion architecture with reliability weighting is fitted on the same data.
4. **Component ablation.** Interaction features, missingness indicators and imputation strategies are individually added and removed to isolate each component's contribution.
5. **Transferability test.** The best configuration, unchanged in architecture, is re-fitted on the secondary type 2 diabetes target, together with that target's own clinical-only and early-fusion counterparts, so that the primary ablation is reproduced in full on the second disease rather than compared across targets.

All configurations share a single fixed experimental protocol described in Section 3.8. No configuration receives more tuning effort, more folds or more data than any other.

3.3 Data Collection Strategy

3.3.1 Data sources

The study uses publicly available, anonymised secondary datasets. Secondary data is appropriate here because the research contribution is methodological rather than epidemiological, because these datasets are established benchmarks that permit comparison with published results, and because their use avoids the ethical and governance burden of primary collection of identifiable health data. Table 3.1 states the datasets selected, their role, and — importantly, given the limitation acknowledged in Section 1.8 — their provenance.

Table 3.1: Datasets selected for the primary and secondary targets

Dataset	Role	Approx. size	Modalities present	Provenance note
Cardiovascular Disease dataset (Kaggle)	Primary CVD training and evaluation	≈70,000 records	Clinical: systolic and diastolic blood pressure, cholesterol category, glucose category, height, weight, age. Lifestyle: smoking, alcohol intake, physical activity	Observational examination records; the only large dataset available to this study containing both modalities in a single table
UCI Heart Disease, Cleveland subset	Primary CVD cross-check	303 records, 14 attributes	Clinical only: resting blood pressure, serum cholesterol, fasting blood sugar, maximum heart rate, ST depression, chest pain type	Detrano *et al.* (1989); long-standing benchmark, but small and clinical-only, so used for external comparability rather than for the fusion experiment
BRFSS-derived Diabetes Health Indicators	Secondary T2D transferability test	≈250,000 records	Lifestyle-dominant: physical activity, fruit and vegetable intake, smoking, heavy alcohol use, general health rating, plus BMI, blood pressure and cholesterol indicators	Derived from the CDC Behavioral Risk Factor Surveillance System (CDC, 2015), the same survey programme used by Xie *et al.* (2019). The commonly distributed class-balanced variant is a balanced *subsample* of real responses, not synthesised data; because subsampling alters the prevalence and therefore the calibration target, results on the balanced variant are reported separately from results on the full-prevalence file
UCI Early Stage Diabetes Risk Prediction	Secondary T2D cross-check	520 records, 17 attributes	Symptom and behavioural indicators without laboratory values	Islam *et al.* (2020); small sample, so reported with confidence intervals and treated as indicative only, per the caution in Section 2.4.1

Dataset	Role	Approx. size	Modalities present	Provenance note
NHANES (continuous cycles)	Real-world missingness regime for RQ4; supplementary hybrid cohort	Cycle-dependent; tens of thousands of records	Full hybrid: examination and laboratory measurements (including HDL cholesterol and HbA1c) alongside an extensive dietary, activity, smoking, alcohol and sleep questionnaire	CDC/NCHS (2024). Included specifically because it carries *naturally occurring* item non-response at realistic rates, which the curated datasets above do not; it is also the source used by Dinh *et al.* (2019), enabling closer external comparison

The addition of NHANES requires explicit justification, because it corrects a weakness that would otherwise undermine two of this project's four claims. The Cardiovascular Disease dataset and the BRFSS-derived Diabetes Health Indicators file are both distributed as cleaned, complete tables with little or no item-level missingness. A study whose contribution includes missingness-aware preprocessing (Novelty 3) and a reliability score derived from behavioural data completeness (Novelty 1) cannot evaluate either claim on data that has no missingness. Restricting the evaluation to artificially deleted values would reduce RQ4 to exactly the synthetic-only assessment that Beretta and Santaniello (2016) warn against, and which Section 2.6.2 makes the centrepiece of its argument. NHANES is therefore included as the real-world missingness regime. On the curated datasets, the only source of genuine missingness is the plausibility recoding of Section 3.4.1, which is reported as a rate rather than assumed to be substantial, and where it proves negligible the reliability score of Section 3.6.2 degenerates to a constant — a possibility that is stated in advance and reported rather than concealed, since a constant reliability score would mean the late-fusion architecture reduces to fixed global weighting on that dataset.

3.3.2 Variable taxonomy

Every variable is assigned to exactly one of three groups before modelling begins, and this assignment is fixed for the duration of the study so that the ablation in Section 3.8 is unambiguous.

- **Group C — Clinical.** Objectively measured or laboratory-derived: systolic and diastolic blood pressure, total and HDL cholesterol, fasting glucose or HbA1c, BMI computed from measured height and weight, resting heart rate.
- **Group L — Lifestyle.** Self-reported behavioural: smoking status and exposure, alcohol consumption, physical activity frequency and intensity, dietary indicators, sleep duration, self-rated general health.
- **Group D — Demographic.** Age and sex, present in both branches as they condition the interpretation of clinical and behavioural variables alike, and used additionally for the fairness analysis of Section 3.11.

This taxonomy is a *schema*, not a claim that every listed variable is present in every dataset, and the difference matters for what can be tested where. The primary Cardiovascular Disease dataset carries blood pressure, cholesterol and glucose as ordinal categories rather than continuous

values, and contains no HDL cholesterol, no HbA1c and no resting heart rate; it carries smoking, alcohol and physical activity as binary indicators, with no dietary or sleep variables. The BRFSS-derived file is the reverse, being rich in behavioural detail and coarse in clinical measurement. Only NHANES contains the full schema. The consequence is that the interaction features of Section 3.5.2 and the branch specifications of Section 3.6.1 are instantiated per dataset from the variables actually available, and Table 3.2 records that availability explicitly. Where a declared interaction cannot be computed on a given dataset, it is reported as not evaluable on that dataset rather than silently omitted.

3.4 Data Preprocessing and the Treatment of Missing Behavioural Data

Preprocessing follows the evidence reviewed in Section 2.6 and is applied strictly inside cross-validation folds — imputation parameters, scaler parameters and any resampling are fitted on training folds only and applied to held-out folds, to avoid the optimistic bias that arises from fitting preprocessing on the full dataset.

3.4.1 Cleaning and physiological plausibility filtering

Duplicate records are removed. Physiologically implausible values — for example systolic blood pressure below 60 mmHg or above 250 mmHg, diastolic exceeding systolic, or BMI outside 12–70 kg/m^2 — are recoded as missing rather than deleted, so that the record's remaining information is preserved and the implausibility is captured by the missingness indicator rather than by row deletion. The number of values recoded at each rule is logged and reported.

3.4.2 Missingness indicators

For every variable in Group L, and for anthropometric variables in Group C, a binary indicator *is_missing* is created and retained as a model feature. This decision is made on the direct evidence of Weng *et al.* (2017), who found that the absence of a recorded BMI value carried predictive signal in its own right. Retaining these indicators preserves informative non-response; it also creates a fairness risk, which is why the subgroup analysis of Section 3.11 is mandatory rather than optional in this design.

3.4.3 Imputation as an experimental variable

Rather than fixing an imputation method by assumption, this study evaluates the following strategies as a controlled factor, with all else held constant:

- **S0 — Complete-case analysis** (reference only, to quantify how much data would be lost).
- **S1 — Mode imputation** for categorical lifestyle variables and median imputation for continuous clinical variables.
- **S2 — KNN imputation** with the neighbourhood size k tuned over {1, 3, 5, 10} on training folds. The inclusion of small k values follows Beretta and Santaniello (2016), whose evaluation found that although larger k reduced imputation error it distorted the data

distribution, and who recommended approximately three neighbours in combination with ReliefF attribute filtering as the best trade-off.

- **S3 — S2 with ReliefF attribute weighting** applied to the distance metric, reproducing the configuration Beretta and Santaniello (2016) identified as optimal.

Each strategy is evaluated on two distinct missingness regimes, following the synthetic-versus-real design of Beretta and Santaniello (2016) and answering RQ4 directly:

1. **Synthetic regime.** Missingness is deleted artificially from complete records at 15% and 30% rates under MCAR and MAR mechanisms, so that ground truth is known and imputation error can be measured directly, alongside distributional fidelity assessed by Kolmogorov–Smirnov distance between the pre- and post-imputation marginal distributions.
2. **Real-world regime.** Naturally occurring missingness is imputed and evaluated indirectly through downstream model performance, since ground truth is unavailable. As set out in Section 3.3.1, this regime is evaluated primarily on NHANES, which carries realistic item non-response; the curated Cardiovascular Disease and BRFSS files are distributed essentially complete, so on those datasets the real-world regime covers only the values recoded by the plausibility rules of Section 3.4.1, and the observed recoding rate is reported so the reader can judge how much weight that arm carries.

Two limitations of this design are recorded rather than left implicit. Beretta and Santaniello (2016) introduced missingness under MCAR only, so the MAR arm of the synthetic regime extends beyond the design their evidence validates. More importantly, missingness not at random (MNAR) — which Section 1.4 argues is the mechanism most likely to govern self-reported behavioural non-response, since patients who decline to report alcohol intake differ systematically from those who do — cannot be simulated with known ground truth at all. It is therefore addressed only indirectly, through the retention of missingness indicators and through the NHANES real-world regime, and this is a genuine limit on the strength of any conclusion drawn under RQ4.

Reporting both regimes matters because a method may minimise imputation error on synthetic MCAR data while distorting the real, non-random missingness structure of self-reported behavioural variables. Divergence between the two regimes is itself a reportable finding.

3.4.4 Scaling, encoding and class imbalance

Continuous features are standardised by Z-score normalisation, fitted on training folds only. Scaling is required for the Support Vector Machine and Logistic Regression branches, since distance- and margin-based learners are sensitive to feature magnitude; tree-based learners receive the unscaled features. Categorical lifestyle variables are one-hot encoded, with ordinal encoding retained where an inherent order exists (for example never / former / current smoker). Class imbalance is addressed by class weighting in the first instance, with SMOTE applied within training folds only as a sensitivity analysis, never to the held-out fold.

3.5 Feature Selection and Engineered Interaction Features

3.5.1 Feature selection

Pearson and point-biserial correlation screening, Chi-square testing for categorical associations, and Recursive Feature Elimination with cross-validation are applied within training folds. Variance inflation factors are inspected for the Logistic Regression branch specifically, since the engineered interaction terms of Section 3.5.2 are by construction correlated with their constituent variables; interaction terms are mean-centred before multiplication to reduce this.

3.5.2 Declared clinical × lifestyle interaction features

Novelty 2 of Section 1.4 is implemented here. The following interaction terms are declared *a priori*, on pathophysiological grounds, rather than being discovered by search. Declaring them in advance is what makes them individually auditable in the SHAP output and separately ablatable in Section 3.8. As Section 2.5.1 explains, they also serve a structural purpose: because late fusion cannot by itself represent cross-modal interactions (Stahlschmidt, Ulfenborg and Synnergren, 2022), these terms are computed before the branch split and supplied to the clinical branch, restoring in explicit form the interaction signal the architecture would otherwise discard. Three of the five are computable on the primary cardiovascular dataset and all five on NHANES, as recorded in the final column of Table 3.2.

Table 3.2: Clinical × lifestyle interaction features and their pathophysiological justification

Interaction term	Justification	Supporting evidence	Computable on
Smoking exposure × systolic blood pressure	Tobacco exposure and raised blood pressure act on the same arterial endothelium; their combined effect on cardiovascular risk is not the sum of their separate effects	Elevated blood pressure alone attributed 25% of global NCD deaths (WHO, 2025); smoking retained as a predictor in Weng *et al.* (2017)	Primary CVD dataset; BRFSS; NHANES
BMI × physical inactivity	Distinguishes elevated BMI accompanied by activity from elevated BMI accompanied by sedentary behaviour, which carry different metabolic risk	Waist size and self-reported weight among top diabetes predictors (Dinh *et al.*, 2019); walking pace informative (Alaa *et al.*, 2019)	Primary CVD dataset; BRFSS; NHANES
Dietary sodium or diet-quality score × blood pressure	Dietary sodium is a direct upstream determinant of blood pressure; the interaction encodes the modifiable pathway rather than only its endpoint	Sodium intake ranked in the top five diabetes predictors (Dinh *et al.*, 2019)	NHANES only (dietary recall); BRFSS proxy via fruit and vegetable intake
Alcohol intake × glucose or HbA1c	Alcohol affects hepatic glucose handling; the joint term captures metabolic risk not visible in either variable alone	Heavy alcohol use retained in BRFSS-based diabetes models (Xie *et al.*, 2019)	NHANES (HbA1c); primary CVD dataset in coarse categorical form
Self-rated general health × age	Self-rated health has different predictive meaning at different ages and acts as a composite proxy for unmeasured burden	Self-reported overall health rating among the most informative of 473 variables (Alaa *et al.*, 2019)	BRFSS; NHANES. Not available on the primary CVD dataset

3.6 Proposed Architecture: Dual-Branch Late Fusion with Reliability Weighting

This section specifies Novelty 1. The architecture is deliberately simple enough to be implemented, tested and explained within the scope of this project, while differing structurally from the early-fusion practice identified in Section 2.5.1.

Figure 3.1: Dual-branch late-fusion architecture with per-record reliability weighting

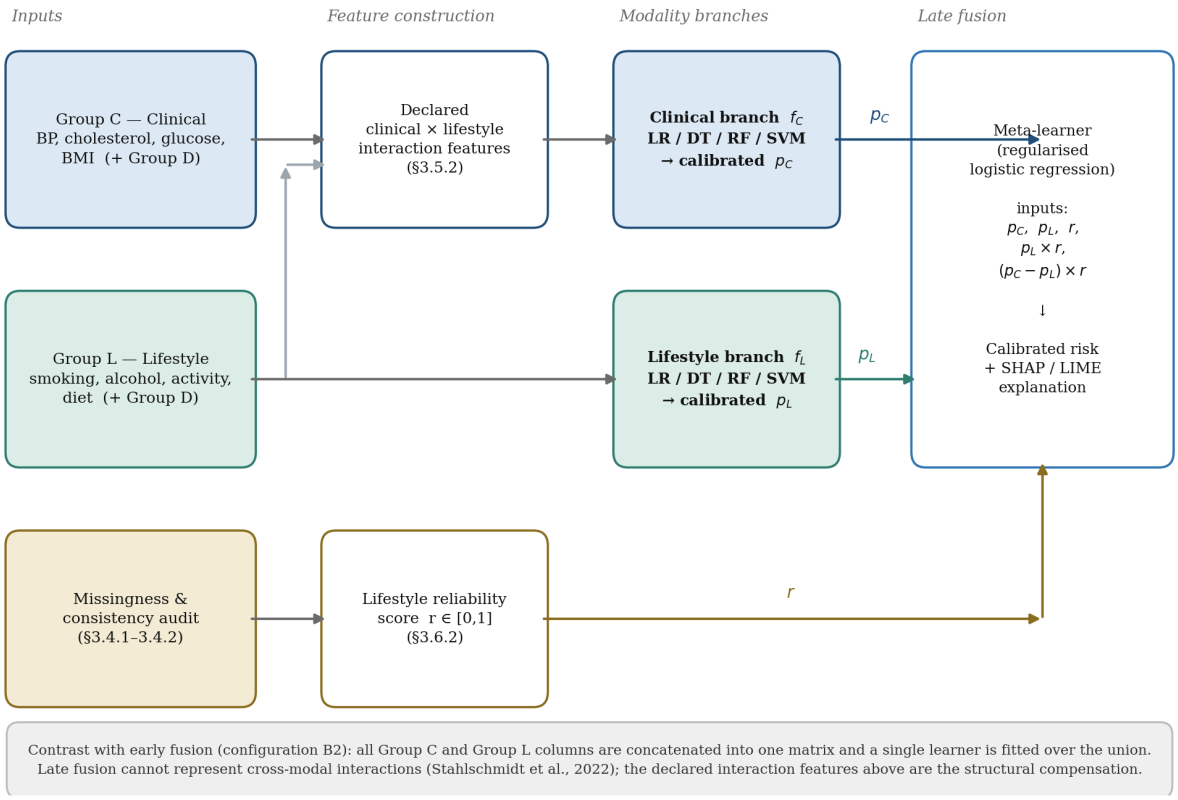


Figure 3.1 — The clinical and lifestyle modalities are modelled by separate calibrated base learners f_C and f_L, whose outputs are combined by a meta-learner together with a per-record lifestyle reliability score r. Declared clinical × lifestyle interaction features are constructed before the branch split and supplied to the clinical branch, reintroducing the cross-modal signal that late fusion would otherwise discard. Configuration B2 in Table 3.3 is the early-fusion alternative against which this structure is tested.

3.6.1 Branch construction

Two base learners are trained independently:

- **Clinical branch f_C**, fitted on Group C features plus Group D demographics, producing a calibrated probability p_C.
- **Lifestyle branch f_L**, fitted on Group L features plus Group D demographics, producing a calibrated probability p_L.

The viability of an independent lifestyle branch is not assumed; it is established by the literature. Xie *et al.* (2019) obtained AUC up to 0.7949 for type 2 diabetes from predominantly behavioural survey variables, and Dinh *et al.* (2019) obtained AUC 0.831 for cardiovascular disease from questionnaire data without laboratory values. Each branch is calibrated separately using Platt scaling or isotonic regression selected on an inner validation split, so that the two

probabilities are on a comparable scale before fusion.

3.6.2 The lifestyle reliability score

For each record a scalar reliability score r [0, 1] is computed from three components, all derived from training-fold statistics only: the proportion of Group L variables observed rather than imputed for that record; an internal-consistency check across logically related behavioural responses (for example, a reported non-drinker with a positive heavy-drinking indicator); and the count of physiologically or logically implausible responses recoded under Section 3.4.1. This score encodes, per patient, how much the behavioural data for that patient can be trusted.

3.6.3 The meta-learner

A regularised Logistic Regression meta-learner is fitted on out-of-fold predictions from both branches, taking as input p_C, p_L, the reliability score r, and the interaction terms $p_L \times r$ and $(p_C - p_L) \times r$. The inclusion of r in interaction with the lifestyle probability is the mechanism by which the fusion weight varies per record: when behavioural data for a patient is sparse or internally inconsistent, the meta-learner can attenuate the lifestyle branch's influence for that patient specifically, rather than the model applying a single global weight learned across all patients. A Logistic Regression meta-learner is chosen deliberately over a more flexible stacker so that the fusion layer itself remains directly inspectable — the fusion weights can be read off the fitted coefficients and reported.

Out-of-fold prediction generation is essential: branch predictions used to train the meta-learner are produced only on data unseen by that branch, preventing the meta-learner from inheriting the branches' training-set optimism.

3.7 Model Development

Four algorithms are used as branch learners, selected on the evidence set out in Section 2.4.1 rather than by convention. Each is tuned by randomised search with a fixed and equal budget of 60 candidate configurations under five-fold inner cross-validation.

- **Logistic Regression.** L1 and L2 regularisation, with the inverse regularisation strength C searched logarithmically. Retained as a demanding baseline on the evidence that it outperformed Random Forest in both Weng *et al.* (2017) and Xie *et al.* (2019).
- **Decision Tree.** Maximum depth, minimum samples per leaf and splitting criterion searched. Retained as an intrinsically interpretable reference point, with the realistic expectation set by Xie *et al.* (2019) that it will be the weakest performer.
- **Random Forest.** Number of estimators, maximum depth, maximum features and minimum samples per split searched. Retained for its native capacity to represent feature interactions, which makes it the natural comparator against the explicitly declared interaction terms of Section 3.5.2.
- **Support Vector Machine.** Linear and RBF kernels both searched, with C and γ tuned and probability outputs obtained by Platt scaling. Both kernels are included because Xie *et al.*

(2019) reported meaningfully different accuracy between linear and RBF variants (80.8% against 81.8%) despite similar AUC.

Data is partitioned by stratified five-fold cross-validation repeated across three random seeds, with a held-out stratified test set of 20% reserved and untouched until final evaluation. All reported results are means with standard deviations across folds and seeds; single-split point estimates are not reported as headline results.

3.8 Benchmark and Ablation Design

This section specifies Novelty 4 and answers the central question of what, exactly, the proposed system is being compared against. Two categories of benchmark are used: *internal* benchmarks, which are the primary evidence, and *external* benchmarks, which situate the results in the published literature.

3.8.1 Internal benchmark configurations

Table 3.3: Benchmark configurations and the comparison each one answers

ID	Configuration	Question it answers	Compared against
B0	Reproduced conventional clinical risk score on the study cohort (see feasibility note below)	How well does current clinical practice perform on this data? (RQ1)	— (floor)
B1	Clinical-only ML model (Group C + D features)	How much does machine learning add over the conventional score, with no lifestyle data at all?	B0
B2	Clinical + lifestyle, early fusion by concatenation	**Primary novelty test.** Does adding the lifestyle modality help, holding the pipeline constant?	**B1**
B3	Clinical + lifestyle, dual-branch late fusion with reliability weighting	Does the proposed fusion strategy beat naive concatenation on identical data?	B2
B4	B3 plus declared interaction features	Do the explicitly engineered interaction terms add value beyond what the ensemble discovers implicitly?	B3
B5	B4 with missingness indicators removed	Is informative non-response contributing to performance?	B4
B6	B4 under each imputation strategy S0–S3	How sensitive is the result to imputation choice, and does that differ between synthetic and real missingness?	B4
B7	Best configuration re-fitted on the secondary T2D target	Do the findings transfer to a second disease?	Its own B1–B2 pair

3.8.2 Feasibility of the conventional-score baseline (B0)

B0 is the floor for every subsequent claim, so its feasibility must be established rather than assumed. The standard instruments — the Framingham risk score (Wilson *et al.*, 1998) and the ACC/AHA pooled cohort equations used as the baseline by Weng *et al.* (2017) — require continuous total and HDL cholesterol, diabetes status and antihypertensive-treatment status. The

primary Cardiovascular Disease dataset carries cholesterol and glucose only as three-level ordinal categories and records neither treatment status nor HDL, so neither instrument can be computed on it exactly. This is stated plainly because a benchmark that cannot be computed is not a benchmark.

Three responses follow, applied in order of preference:

1. **NHANES is the primary site for B0.** NHANES carries the continuous laboratory values, treatment status and diabetes status that Framingham requires, so the conventional score is reproduced exactly there, and B0–B1–B2 are run as a complete set on that cohort.
2. **On the primary Cardiovascular Disease dataset, a documented approximation is used and labelled as such.** Category midpoints are substituted for the ordinal cholesterol and glucose levels and treatment status is set to its cohort-modal value; the resulting score is reported as an *approximated* Framingham baseline, never as the instrument itself, and its sensitivity to the midpoint choice is reported.
3. **B1 is the operative floor where B0 cannot be computed faithfully.** The primary novelty test is B1 against B2, and that comparison does not depend on B0. B0 provides clinical context, not the evidentiary base, and where it is only approximable the conclusions of this study rest on B1.

3.8.3 The control conditions that make the comparison valid

The comparison **B1 against B2** is the primary test of this project's central claim, and its validity depends entirely on the two configurations differing in one respect only. The following are therefore held identical across every configuration: the cross-validation fold assignments; the random seeds; the hyperparameter search space and the number of search iterations; the class-balancing strategy; the calibration procedure; the decision threshold selection rule; and the held-out test set. Only the feature set differs. Any performance difference is therefore attributable to the lifestyle modality and not to incidental pipeline variation — which is precisely the confound that comparisons against published results from other pipelines cannot exclude.

One caveat applies to the meta-learner of Section 3.6.3, since stacked models are a well-known site of information leakage. The branch predictions used to fit the meta-learner are generated by an *inner* five-fold cross-validation nested strictly within each outer training fold, so that no record contributes to both a branch's training and the meta-learner's view of that branch's behaviour on it. The outer fold assignments, which Section 3.8.3 requires to be identical across configurations, are unaffected by this inner loop. Configurations B1 and B2, which contain no meta-learner, incur the same inner cross-validation cost so that the tuning budget is genuinely equal across configurations rather than merely nominally so.

3.8.4 Statistical testing

Differences in AUC between paired configurations are tested using DeLong's test for correlated ROC curves, since the models are evaluated on the same samples. Differences in classification decisions are tested using McNemar's test. Confidence intervals for all reported metrics are

obtained by bootstrap resampling of the test set with 2,000 replicates. A difference in point estimates is not reported as an improvement unless it is accompanied by a significance test and an effect size, and the direction of any non-significant difference is reported honestly rather than omitted.

Because this design generates a substantial number of comparisons — eight benchmark configurations, four imputation strategies, two missingness regimes and stratified subgroup analyses — multiplicity is controlled explicitly rather than ignored. A single comparison, **B1 against B2 on the primary cardiovascular dataset**, is designated the confirmatory test and is evaluated at $\alpha = 0.05$ without adjustment. Every other comparison is designated exploratory, and *p*-values across the exploratory family are adjusted by the Benjamini–Hochberg procedure at a false discovery rate of 0.05. This allocation, the fold seeds and the primary and secondary metrics are fixed and recorded in the project repository before any model is fitted, so that the confirmatory test cannot be reassigned after the results are seen.

3.8.5 External benchmarks

External results are used chiefly for context, since differences in cohort, outcome definition and evaluation protocol make direct comparison unsafe in most cases. One exception is noted: Paul, Ibrahim and Masood (2026) report on an approximately 70,000-record cardiovascular dataset assembled from the same Cardio and UCI sources used here, so their figures are the one external reference that can be compared to this study's primary results on something close to like-for-like data. Their result is therefore treated as a target to be matched or exceeded rather than merely as background, while acknowledging that their preprocessing and split protocol are not identical to this study's and that any comparison remains indicative. The published reference points identified in Chapter 2 are: ACC/AHA baseline AUC 0.728 and best machine learning AUC 0.764 on 378,256 primary-care patients (Weng *et al.*, 2017); Framingham AUC 0.724 and automated machine learning AUC 0.774 on 423,604 UK Biobank participants (Alaa *et al.*, 2019); cardiovascular AUC 0.831 without laboratory data and 0.839 with, and diabetes AUC 0.862 and 0.957 respectively, on NHANES (Dinh *et al.*, 2019); the full six-model comparison on 2014 BRFSS data reported by Xie *et al.* (2019); and AUC 0.773–0.794 on a comparable ≈70,000-record cardiovascular dataset (Paul, Ibrahim and Masood, 2026). The last of these is the closest external comparator, for the reason given above.

3.9 Evaluation Metrics

Accuracy alone is inadequate for this task, both because the classes are imbalanced and because the clinical costs of the two error types are grossly asymmetric: a false positive leads to further testing and avoidable anxiety, whereas a false negative leads to an undetected condition progressing untreated.

Table 3.4: Evaluation metrics and their clinical interpretation

Metric	Definition	Clinical interpretation and priority
Recall (Sensitivity)	Proportion of true cases correctly identified	**Primary.** Directly measures the failure mode this project exists to reduce — missed early cases
ROC-AUC	Threshold-independent discrimination	**Primary.** The comparable quantity across all internal and external benchmarks
PR-AUC	Area under the precision-recall curve	More informative than ROC-AUC under class imbalance; reported alongside it
Precision	Proportion of positive predictions that are correct	Governs the burden of unnecessary follow-up testing placed on services and patients
F1-score	Harmonic mean of precision and recall	Balanced summary for comparison with published studies that report it
Accuracy	Overall proportion correct	Reported for comparability with the literature only; never used as the deciding metric
Calibration (Brier score, calibration curve)	Agreement between predicted probabilities and observed frequencies	**Primary.** A risk score that is well ranked but poorly calibrated cannot support a clinical threshold decision, and calibration is a precondition for the fusion layer of Section 3.6
Subgroup metrics	All of the above, stratified by age band and sex	Detects performance disparities masked by aggregate figures (Section 3.11)

3.10 Integration of Explainable AI

SHAP and LIME are assigned distinct, non-overlapping roles, following the analysis in Section 2.7.4.

3.10.1 SHAP: global model audit

SHAP (Lundberg and Lee, 2017) is applied to produce the global feature hierarchy across the population, using TreeSHAP for the tree-based branches for exact and efficient computation (Lundberg *et al.*, 2020) and KernelSHAP for the SVM branch. Summary and beeswarm plots establish which clinical and lifestyle factors drive risk across the cohort and in which direction; dependence plots examine whether the model has learned the interaction structure that the declared features of Section 3.5.2 were designed to encode. SHAP is also used to inspect whether missingness indicators are carrying disproportionate weight, which feeds directly into the fairness analysis.

3.10.2 LIME: per-patient explanation at the point of care

LIME (Ribeiro, Singh and Guestrin, 2016) is applied at the level of the individual prediction to generate the sparse, three-to-five-feature explanation that a clinician would see alongside a risk estimate during a consultation. Its specific role here is to identify the *modifiable* features to which the model is locally most sensitive for that patient, together with the direction and magnitude of that local sensitivity. This is not the same as identifying the minimal change that would reclassify the patient — as Section 2.7.4 notes, that is counterfactual explanation and is

not what LIME computes — but a signed local sensitivity over behavioural features is the information from which a clinician can judge which lifestyle factors the model is currently reacting to, and it is not directly provided by a population-relative SHAP attribution. The number of features surfaced is fixed at five and the neighbourhood kernel width is tuned on a validation split rather than left at the library default, since LIME explanations are known to be sensitive to both.

3.10.3 Measured agreement between the two methods

Because Ghassemi, Oakden-Rayner and Beam (2021) argue that post-hoc explanations may be plausible without being faithful, explanation agreement is treated as a measurable outcome rather than an assumption. For a stratified sample of test-set predictions, the top-*k* feature sets produced by SHAP and by LIME are compared using rank correlation and Jaccard overlap, and sign agreement on the direction of each shared feature's contribution is recorded. Stability is assessed by repeating LIME with different perturbation seeds. Cases of systematic disagreement are reported as findings, not suppressed, and are cross-checked against the intrinsically interpretable Decision Tree of Section 3.7.

3.11 Ethics, Data Governance and Fairness

All data used is publicly available, anonymised and licensed for research use; no identifiable patient data is processed and no attempt at re-identification is made. Data is stored in the university-approved environment and processed in compliance with the UK GDPR and the Data Protection Act 2018. Ethical approval will be sought through the standard institutional process for research using secondary anonymised data.

Fairness is treated as a required analysis rather than a statement of intent. Recall, precision, ROC-AUC and calibration are reported stratified by sex and by age band. Because Section 3.4.2 deliberately retains missingness indicators as features, and because behavioural non-response is known to vary with socio-economic position, the SHAP analysis of Section 3.10.1 is used specifically to test whether the model is placing substantial weight on non-response patterns. Where the recall of any reported subgroup falls more than **five percentage points** below the pooled recall — a threshold fixed in advance alongside the analysis plan of Section 3.8.4, not chosen after inspecting results — the disparity is reported explicitly, its likely mechanism is discussed, and the SHAP attribution for the affected subgroup is examined for disproportionate reliance on missingness indicators. Subgroups with fewer than 200 records in the test set are reported with their confidence intervals but excluded from the threshold rule, since the estimate is too unstable to act on.

3.12 Implementation Environment

The system is implemented in Python 3.11 using scikit-learn for modelling, preprocessing and cross-validation; pandas and NumPy for data handling; imbalanced-learn for resampling; the *shap* and *lime* packages for the explanation layer; and matplotlib and seaborn for visualisation. All random seeds are fixed and recorded, environment versions are pinned, and the full

experimental configuration for each benchmark in Table 3.3 is stored so that every reported result is reproducible.

3.13 Project Timeline

Table 3.5: Project timeline

Phase	Activity	Deliverable	Duration
Phase 1	Dataset acquisition, licence verification, exploratory analysis, missingness characterisation	Data audit report	Weeks 1–2
Phase 2	Preprocessing pipeline; missingness indicators; imputation experiments S0–S3 on synthetic and real regimes	Reproducible pipeline; RQ4 results	Weeks 3–5
Phase 3	Feature engineering and declared interaction terms; feature selection	Documented feature set	Week 6
Phase 4	Baseline models B0–B2; hyperparameter tuning; primary ablation with DeLong testing	RQ1, RQ2 and RQ5 results	Weeks 7–9
Phase 5	Late-fusion architecture, reliability scoring, meta-learner; configurations B3–B6	RQ3 results	Weeks 10–12
Phase 6	SHAP and LIME integration; agreement analysis; fairness and subgroup analysis	RQ6 results; fairness report	Weeks 13–14
Phase 7	Transferability test on the secondary type 2 diabetes target (B7)	Generalisation evidence	Week 15
Phase 8	Results write-up, discussion against external benchmarks, conclusion and submission	Final dissertation	Weeks 16–18

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Appendix A: Response to First-Review Supervisory Feedback

This appendix maps each point of the supervisory feedback received on 10 August to the specific location in this revised report where it has been addressed. It is included so that the changes made between versions can be verified directly.

#	Feedback point	Addressed in	What was changed
1.1	**Include specific numbers and scale.** Quantify the problem with WHO figures, NHS costs, or percentage of late-stage diagnoses.	§1.1, Table 1.1; **§1.2**; §2.2	Section 1.1 was rewritten around cited primary-source figures: 43 million NCD deaths and at least 19 million CVD deaths in 2021 with 25% of NCD deaths attributed to elevated blood pressure (WHO, 2025); £29.02 bn UK CVD burden with a £16.62 bn direct-cost breakdown (Shih *et al.*, 2025); ≈£14 bn cost of diabetes with >£10 bn falling on the NHS (Hex *et al.*, 2024); 41.3% of staged common cancers still diagnosed at stage 3 or 4 (NHS England, 2025); and 1.3 million people with undiagnosed type 2 diabetes plus 6.9 million at increased risk (Diabetes UK, 2025). Table 1.1 consolidates all figures with sources. **Section 1.2 was also quantified, as the feedback specified:** its first pillar now carries the late-stage and undiagnosed figures, and its second pillar carries new figures on the measured discriminative ceiling of conventional risk scores — ACC/AHA at AUC 0.728 across 378,256 patients (Weng *et al.*, 2017) and Framingham at 0.724 across 423,604 participants (Alaa *et al.*, 2019).
1.2	**Clarify scope.** State whether the pipeline addresses a specific target disease or acts as a generalised multi-disease framework.	§1.3, Table 1.2	A new dedicated section states the position explicitly: cardiovascular disease is the *primary* target, type 2 diabetes is a *secondary transferability test*, and a generalised multi-disease classifier is explicitly disclaimed. Cancer is excluded as a modelling target with a stated reason. Table 1.2 lists in-scope and out-of-scope items across five dimensions. The report title has been narrowed accordingly.
2.1	**Beyond "combining X and Y".** Merging tabular lifestyle and clinical data is already widely practised; articulate how the integration strategy differs — feature engineering, a specific fusion technique, or different handling of missing/noisy behavioural data.	§1.4; §2.5.1; §3.5.2; §3.6; §3.4	Section 1.4 opens by conceding that hybrid modelling is already common, citing the studies that do it, and then states four specific, falsifiable claims: (1) dual-branch *late fusion* with a per-record lifestyle reliability weight, replacing early fusion by concatenation; (2) declared clinical × lifestyle interaction features with pathophysiological justification; (3) missingness-aware handling of self-reported behavioural data with imputation treated as an experimental variable; (4) a pipeline-controlled ablation as the evidentiary standard. Each is specified operationally in Chapter 3 (§3.6, §3.5.2, §3.4, §3.8 respectively). A new Section 2.5.1 reviews the multimodal fusion literature itself (Huang *et al.*, 2020; Stahlschmidt, Ulfenborg and Synnergren, 2022), including the finding that no fusion strategy dominates across domains and the objection that late fusion cannot represent cross-modal interactions — which is why the interaction features of Novelty 2 are computed before the branch split as structural compensation. Section 2.5.2 identifies the gap that motivates late fusion — that the marginal value of laboratory over questionnaire data was 0.8 AUC points for CVD but 9.5 points for diabetes in Dinh *et al.* (2019), a difference a fixed feature matrix cannot represent.

#	Feedback point	Addressed in	What was changed
2.2	**Benchmark comparison.** State what existing baseline the hybrid model is compared against. Is novelty proven by comparing a clinical-only model directly against a clinical + lifestyle model using the exact same pipeline?	§3.8, Table 3.3; §3.8.2–3.8.4	Yes, and this is now the primary test. Section 3.8 defines eight benchmark configurations (B0–B7); the comparison **B1 (clinical-only) against B2 (clinical + lifestyle)** is designated the primary novelty test. Section 3.8.2 lists the control conditions that make it valid — identical folds, seeds, hyperparameter search budget, class balancing, calibration, threshold rule and held-out test set, with only the feature set differing. Section 3.8.3 adds DeLong testing for paired AUCs, McNemar testing for decisions and bootstrap confidence intervals. Section 3.8.4 lists the external published baselines used for context (Weng *et al.*, 2017; Alaa *et al.*, 2019; Dinh *et al.*, 2019; Xie *et al.*, 2019; Paul, Ibrahim and Masood, 2026).
3.1	**Literature review as methodological foundation.** Chapter 2 should not read like textbook definitions; it must review prior studies applying these models to similar hybrid healthcare datasets, show what performance previous researchers achieved, and explain how that justifies the choice of baseline algorithms.	§2.1; §2.4, Table 2.1; §2.4.1; §2.5, Table 2.2	Chapter 2 was restructured entirely. The textbook definitions of Logistic Regression, Decision Trees, Random Forests and SVM have been removed and replaced by an evidence review. Table 2.1 reports seven prior studies with their datasets, feature modalities and reported performance (numeric per-algorithm figures where the original papers published them; two rows, marked †, are clinical-only studies included for algorithmic rather than modality evidence). Section 2.4.1 then derives the algorithm selection from that evidence: Logistic Regression is retained because it *beat* Random Forest in Weng *et al.* (2017) (AUC 0.760 vs 0.745) and in Xie *et al.* (2019) (0.7932 vs 0.7608), so it is a demanding baseline; both SVM kernels are searched because Xie *et al.* reported diverging accuracy between them (80.8% vs 81.8%); Random Forest is retained for native interaction modelling; and the Decision Tree is retained as an interpretability reference with the realistic expectation, from Xie *et al.*'s AUC of 0.7182, that it will be weakest. Section 2.1 also documents the review's search strategy.
3.2	**Deepen technological literature context.** If KNN imputation is to be used for missing lifestyle values, include prior literature in Chapter 2 evaluating how KNN or mode imputation affects synthetic versus real-world clinical datasets.	§2.6, Table 2.3; §3.4.3	A new Section 2.6 reviews this literature directly. Jerez *et al.* (2010) is cited for the finding that machine learning imputation (including KNN) significantly outperformed mean/mode, hot-deck and multiple imputation on real breast cancer data. Beretta and Santaniello (2016) is reviewed in detail because its design is exactly the synthetic-versus-real comparison requested: 1NN, KNN (k=5) and weighted KNN across synthetic data at 15% and 30% missingness and the real SPECTF heart dataset at 15% MCAR, with the finding that larger k reduces imputation error but distorts the data distribution, and that ≈3 neighbours with ReliefF filtering gives the best trade-off. Section 3.4.3 implements this as an experiment (strategies S0–S3, k tuned over {1, 3, 5, 10}, evaluated on both synthetic and real missingness regimes with distributional fidelity measured), which is now research question RQ4.

#	Feedback point	Addressed in	What was changed
3.3	**Differentiate SHAP vs LIME.** In Section 2.6.2, explain why *both* are necessary rather than choosing one. What complementary clinical insight does LIME's local surrogate approach offer over SHAP's global game-theoretic approach in a live decision-support setting?	§2.7.2–2.7.5, Table 2.4; §3.10	Section 2.7.4 answers this directly on four grounds. (i) *Explanandum*: SHAP attributes a prediction relative to a population background — "why is this patient higher risk than average?" — whereas LIME approximates the local decision boundary — "what is the smallest change that would move this patient out of the high-risk band?", which is the question that drives an intervention conversation. (ii) *Cost at the point of care*: TreeSHAP is exact and fast only for tree models, so the SVM branch requires slow model-agnostic estimation, while LIME's sparse local regression returns an explanation within interactive latency. (iii) *Output form*: SHAP is dense, additive and suited to audit; LIME is sparse and rule-like, matching how clinical reasoning is documented. (iv) *Disagreement as diagnostic signal*: because the two rest on different foundations, their agreement is convergent evidence and their disagreement is informative, so Section 3.10.3 measures it via rank correlation, Jaccard overlap and sign agreement rather than assuming either is correct. Table 2.4 tabulates the comparison, and Section 2.7.5 adds the necessary critical counterweight from Ghassemi, Oakden-Rayner and Beam (2021) on the limits of post-hoc explanation.